

AI Agent Formulation and Analysis

Selected Agent: AI Study Assistant

1. Problem in the Agent's Context

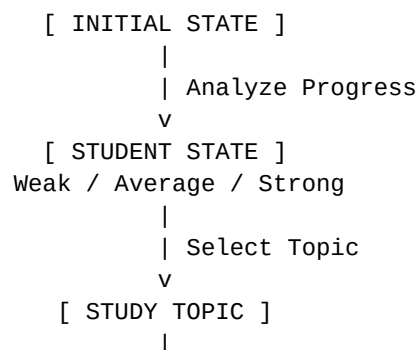
The main problem is to help a student study effectively and achieve their learning goals. The agent needs to understand the student's subjects, completed and uncompleted topics, quiz performance, weak areas, and available study time. Based on this information, the agent recommends suitable study topics, revision, practice questions, quizzes, and study plans.

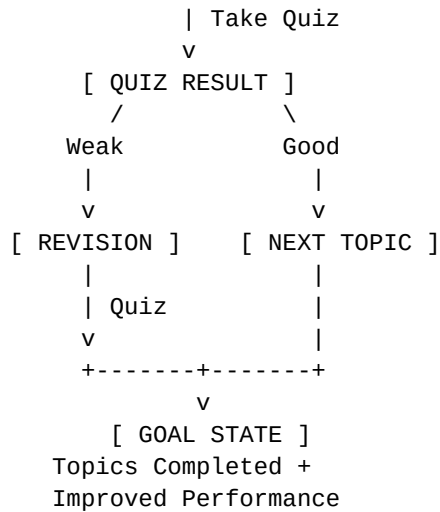
2. Problem Formulation

Textual Formulation

- **Problem Type:** Multiple-State Problem (Sensorless / Uncertain State) because the agent does not initially know the student's exact level of understanding of different topics.
- **States:** The student's current learning condition, including their subjects, completed and uncompleted topics, quiz scores, weak topics, and available study time.
- **Initial State:** The student provides their subjects, current progress, previous quiz scores, weak topics, and available study time. The agent has uncertain knowledge about the student's actual understanding.
- **Actions:** Recommend Topic, Recommend Revision, Generate Practice Questions, Generate Quiz, Create Study Plan, Update Study Plan.
- **Transition Model:** Studying a topic changes the student's progress; completing a quiz provides a new performance result; a weak result may change the student's state to require revision, while a good result may allow the agent to recommend the next topic.
- **Goal Test:** The required topics are completed and the student achieves the desired learning or quiz performance.
- **Path Cost:** Study time and unnecessary or repeated study actions have a cost. The goal is to achieve the required learning outcome using the available study time efficiently.

Graphical Formulation





3. PEAS Evaluation

Performance Measure	Improvement in quiz scores, completion of study goals, efficient use of study time, and reduction of weak topics.
Environment	Student, subjects, topics, study materials, quizzes, assignments, and study schedule.
Actuators	Study-plan recommendations, topic recommendations, quizzes, practice questions, revision suggestions, and explanations.
Sensors	Quiz scores, student responses, completed topics, study progress, and student feedback.

4. Environment Properties

- Partially Observable
- Stochastic
- Sequential
- Dynamic
- Discrete
- Single-Agent